

2 Unit-I: Groups

Definition 2.1 (Group). A **group** is a nonempty set G together with a binary operation $*$ such that, for all $a, b, c \in G$, the following axioms hold:

1. **Closure:** $a * b \in G$.
2. **Associativity:** $(a * b) * c = a * (b * c)$.
3. **Identity element:** There exists an element $e \in G$ such that $e * a = a * e = a$ for every $a \in G$.
4. **Inverse element:** For every $a \in G$, there exists an element $a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$.

The group is denoted by $(G, *)$.

If, in addition,

5. $a * b = b * a$, for all $a, b \in G$,

then $(G, *)$ is called an **abelian group** or a **commutative group**.

(Question 1 & 2-QB)

Example 2.1.

1. The set $(\mathbb{Z}, +)$ of Integers is an abelian group under usual addition ‘+’.
2. The set $(\mathbb{Q}^\times = \mathbb{Q} \setminus \{0\}, \cdot)$ of non-zero Rational numbers is an abelian group under usual multiplication ‘ $\cdot$ ’.
3. The set of Natural numbers $\mathbb{N} = \{1, 2, 3, \dots\}$ under usual addition ‘+’ is **not** a group.

(Question 1 & 2-QB)

Problem 2.1. If $\mathbb{Q}_0$ is the set of all rational numbers other than zero and $*$ is a binary operation on $\mathbb{Q}_0$ defined by $a * b = \frac{ab}{5}$, $\forall a, b \in \mathbb{Q}_0$, then prove that $(\mathbb{Q}_0, *)$ forms an abelian group.

(Question 21-QB)

Solution

1. Closure:

Let $a, b \in \mathbb{Q}_0$. Since a and b are nonzero rational numbers, $ab \neq 0$. Also, $\frac{ab}{5} \in \mathbb{Q}$ and $\frac{ab}{5} \neq 0$. Therefore, $a * b = \frac{ab}{5} \in \mathbb{Q}_0$.

Hence, $\mathbb{Q}_0$ is closed under $*$.

2. Associativity:

Let $a, b, c \in \mathbb{Q}_0$. Then $(a * b) * c = \left(\frac{ab}{5}\right) * c$.

Using the definition of $*$, we have $(a * b) * c = \frac{\left(\frac{ab}{5}\right) c}{5} = \frac{abc}{25}$.

$$\text{Similarly, } a * (b * c) = a * \left(\frac{bc}{5}\right) = \frac{a\left(\frac{bc}{5}\right)}{5} = \frac{abc}{25}.$$

Thus, $(a * b) * c = a * (b * c)$.

Therefore, $*$ is associative.

3. Identity:

Let $e \in \mathbb{Q}_0$ be the identity element. Then $a * e = a$ for every $a \in \mathbb{Q}_0$.

By the definition of $*$, we have $\frac{ae}{5} = a \implies ae = 5a$.

Since $a \neq 0$, $e = 5$.

Also, $a * 5 = \frac{5a}{5} = a$ and $5 * a = \frac{5a}{5} = a$.

Hence, the identity element is $\boxed{e = 5}$.

4. Inverse:

Let $a \in \mathbb{Q}_0$. We seek $a^{-1} \in \mathbb{Q}_0$ such that

$a * a^{-1} = 5$, where 5 is the identity element.

Thus, $\frac{a a^{-1}}{5} = 5$.

Therefore, $a a^{-1} = 25$, and hence $a^{-1} = \frac{25}{a}$.

Since $a \neq 0$, $\frac{25}{a} \in \mathbb{Q}_0$.

Moreover, $a * \frac{25}{a} = \frac{a\left(\frac{25}{a}\right)}{5} = \frac{25}{5} = 5$.

Similarly, $\frac{25}{a} * a = 5$.

Hence every element $a \in \mathbb{Q}_0$ has an inverse given by $\boxed{a^{-1} = \frac{25}{a}}$.

5. Commutativity:

For $a, b \in \mathbb{Q}_0$, it is given that $a * b = \frac{ab}{5}$.

Since ordinary multiplication of rational numbers is commutative, $ab = ba$.

Therefore, $a * b = \frac{ab}{5} = \frac{ba}{5} = b * a$.

Hence $*$ is commutative.

Therefore,

$$\boxed{(\mathbb{Q}_0, *) \text{ is an abelian group.}}$$

Problem 2.2. Show that the set $\mathbb{Q}_{-1}$ of all rational numbers except -1 , with operation defined by $a * b = a + b + ab$, $\forall a, b \in \mathbb{Q}_{-1}$, is an abelian group.

(Question 22-QB)

Solution

1. Closure:

Let $a, b \in \mathbb{Q}_{-1}$. Since $a, b \in \mathbb{Q}$, $a + b + ab \in \mathbb{Q}$.

It remains to show that $a + b + ab \neq -1$.

Suppose, for contradiction, that $a + b + ab = -1$.

Then $ab + a + b + 1 = 0$.

Factoring, we get $(a + 1)(b + 1) = 0$.

Therefore, $a + 1 = 0$ or $b + 1 = 0$, which gives $a = -1$ or $b = -1$.

But $a, b \in \mathbb{Q}_{-1}$, so neither a nor b is equal to -1 . This is a contradiction.

Hence, $a * b = a + b + ab \neq -1$.

Therefore, $a * b \in \mathbb{Q}_{-1}$.

Hence, $\mathbb{Q}_{-1}$ is closed under $*$.

2. Associativity:

Let $a, b, c \in \mathbb{Q}_{-1}$. Then $(a * b) * c = (a + b + ab) * c$.

Using the definition of $*$, we have $(a * b) * c = (a + b + ab) + c + (a + b + ab)c$.

Therefore, $(a * b) * c = a + b + c + ab + ac + bc + abc$.

Similarly, $a * (b * c) = a * (b + c + bc)$.

Using the definition of $*$, we have $a * (b * c) = a + (b + c + bc) + a(b + c + bc)$.

Therefore, $a * (b * c) = a + b + c + bc + ab + ac + abc = a + b + c + ab + ac + bc + abc$.

Thus, $(a * b) * c = a * (b * c)$.

Therefore, $*$ is associative.

3. Identity:

Let $e \in \mathbb{Q}_{-1}$ be the identity element. Then $a * e = a$ for every $a \in \mathbb{Q}_{-1}$.

By the definition of $*$, we have $a + e + ae = a$.

Therefore, $e + ae = 0$, or $e(1 + a) = 0$.

Since $a \neq -1$, $1 + a \neq 0$.

Hence, $e = 0$.

Also, $a * 0 = a + 0 + a(0) = a$ and $0 * a = 0 + a + 0(a) = a$.

Hence, the identity element is $\boxed{e = 0}$.

4. Inverse:

Let $a \in \mathbb{Q}_{-1}$. We seek $a^{-1} \in \mathbb{Q}_{-1}$ such that

$a * a^{-1} = 0$, where 0 is the identity element.

Thus, $a + a^{-1} + aa^{-1} = 0 \implies a^{-1}(1 + a) = -a$.

Since $a \neq -1$, $1 + a \neq 0$.

Hence, $a^{-1} = -\frac{a}{1 + a}$.

Since $a \in \mathbb{Q}$ and $a \neq -1$, $-\frac{a}{1 + a} \in \mathbb{Q}$.

We must also show that $-\frac{a}{1 + a} \neq -1$.

Suppose $-\frac{a}{1 + a} = -1$.

Then $-a = -(1 + a)$, which implies $-a = -1 - a$, and hence $0 = -1$, which is impossible.

Therefore, $-\frac{a}{1 + a} \neq -1$.

Thus, $-\frac{a}{1 + a} \in \mathbb{Q}_{-1}$.

Moreover, $a * \left(-\frac{a}{1 + a}\right) = a - \frac{a}{1 + a} - a\frac{a}{1 + a}$.

Therefore, $a * \left(-\frac{a}{1 + a}\right) = \frac{a(1 + a) - a - a^2}{1 + a} = 0$.

Similarly, $\left(-\frac{a}{1 + a}\right) * a = 0$.

Hence every element $a \in \mathbb{Q}_{-1}$ has an inverse given by $a^{-1} = -\frac{a}{1+a}$.

5. Commutativity:

For $a, b \in \mathbb{Q}_{-1}$, it is given that $a * b = a + b + ab$.

Since ordinary addition and multiplication of rational numbers are commutative, we have $a + b = b + a$ and $ab = ba$.

Therefore, $a * b = a + b + ab = b + a + ba = b * a$.

Hence $*$ is commutative.

Therefore,

$$(\mathbb{Q}_{-1}, *) \text{ is an abelian group.}$$

General Properties of Groups

Note. For simplicity, throughout these notes we often write ab instead of $a * b$. Here, ab denotes the group operation $*$ and does not necessarily represent ordinary multiplication.

Problem 2.3. Prove that identity element in a group is unique.

(Question 4-QB)

Solution

Suppose that e and e' are two identity elements of G . Then

$$\begin{aligned} e &= ee' & (\because e' \text{ is an identity element}), \\ ee' &= e' & (\because e \text{ is an identity element}). \end{aligned}$$

Hence $e = e'$. Therefore the identity element of a group is unique.

Problem 2.4. Prove that inverse of every element in a group is unique.

(Question 5-QB)

Solution

Let $a \in G$, and suppose that both b and c are inverses of a . Then by the inverse axiom

$$ab = ba = e \quad \text{and} \quad ac = ca = e.$$

Now,

$$\begin{aligned}
 b &= be & (\because \text{identity axiom}) \\
 &= b(ac) & (\because ac = e, \text{ inverse axiom}) \\
 &= (ba)c & (\because \text{associativity}) \\
 &= ec & (\because ba = e, \text{ inverse axiom}) \\
 &= c & (\because \text{identity axiom}).
 \end{aligned}$$

Thus $b = c$. Hence the inverse of a is unique.

Problem 2.5. Prove that in a group $(G, *)$, $(a^{-1})^{-1} = a$, for every $a \in G$.

(Question 6-QB)

Solution

By the inverse axiom,

$$a^{-1}a = e.$$

Therefore,

$$\begin{aligned}
 (a^{-1})^{-1}(a^{-1}a) &= (a^{-1})^{-1}e & (\because \text{multiplying both sides on the left by } (a^{-1})^{-1}) \\
 \left[(a^{-1})^{-1}a^{-1}\right]a &= (a^{-1})^{-1} & (\because \text{associativity and identity axiom}) \\
 (e)a &= (a^{-1})^{-1} & (\because \text{since } (a^{-1})^{-1} \text{ is the inverse of } a^{-1}) \\
 a &= (a^{-1})^{-1} & (\because \text{identity axiom}).
 \end{aligned}$$

Hence,

$$\boxed{(a^{-1})^{-1} = a}.$$

Problem 2.6. Let G be a group. Then for all $a, b \in G$, prove that $(ab)^{-1} = b^{-1}a^{-1}$. That is, prove that the inverse of a product is the product of the inverses taken in the reverse order.

(Question 28-QB)

Solution

Let $a, b \in G$. We show that $b^{-1}a^{-1}$ is both a right and a left inverse of ab .

First,

$$\begin{aligned}
 (ab)(b^{-1}a^{-1}) &= a(bb^{-1})a^{-1} & (\because \text{associativity}) \\
 &= aea^{-1} & (\because bb^{-1} = e, \text{ inverse axiom}) \\
 &= aa^{-1} & (\because \text{identity axiom}) \\
 &= e & (\because \text{inverse axiom}).
 \end{aligned}$$

Also,

$$\begin{aligned}
 (b^{-1}a^{-1})(ab) &= b^{-1}(a^{-1}a)b && (\because \text{associativity}) \\
 &= b^{-1}eb && (\because a^{-1}a = e, \text{ inverse axiom}) \\
 &= b^{-1}b && (\because \text{identity axiom}) \\
 &= e && (\because \text{inverse axiom}).
 \end{aligned}$$

Hence $b^{-1}a^{-1}$ is the inverse of ab . By uniqueness of inverse,

$$\boxed{(ab)^{-1} = b^{-1}a^{-1}}.$$

Problem 2.7. Let G be a group and $a, b, c \in G$. Then prove that

$$ab = ac \implies b = c \quad (\text{left cancellation law}),$$

and

$$ba = ca \implies b = c \quad (\text{right cancellation law}).$$

(Question 8-QB)

Solution

Suppose first that $ab = ac$. Multiplying both sides on the left by a^{-1} , we get

$$\begin{aligned}
 a^{-1}(ab) &= a^{-1}(ac) && (\because \text{equality is preserved under left multiplication}) \\
 (a^{-1}a)b &= (a^{-1}a)c && (\because \text{associativity}) \\
 eb &= ec && (\because a^{-1}a = e, \text{ inverse axiom}) \\
 b &= c && (\because \text{identity axiom}).
 \end{aligned}$$

Thus the left cancellation law holds.

Now suppose that $ba = ca$. Multiplying both sides on the right by a^{-1} , we obtain

$$\begin{aligned}
 (ba)a^{-1} &= (ca)a^{-1} && (\because \text{equality is preserved under right multiplication}) \\
 b(aa^{-1}) &= c(aa^{-1}) && (\because \text{associativity}) \\
 be &= ce && (\because aa^{-1} = e, \text{ inverse axiom}) \\
 b &= c && (\because \text{identity axiom}).
 \end{aligned}$$

Hence the right cancellation law also holds.

Problem 2.8. If $a, b \in G$, then prove that each of the equations

$$ax = b \quad \text{and} \quad ya = b$$

has a unique solution in G .

(Question 29-QB)

Solution

Consider first the equation $ax = b$. Since $a^{-1}, b \in G$, closure gives $a^{-1}b \in G$. Put $x = a^{-1}b$. Then

$$\begin{aligned} ax &= a(a^{-1}b) \\ &= (aa^{-1})b && (\because \text{associativity}) \\ &= eb && (\because aa^{-1} = e, \text{ inverse axiom}) \\ &= b && (\because \text{identity axiom}). \end{aligned}$$

Thus $x = a^{-1}b$ is a solution.

To prove uniqueness, suppose x_1 and x_2 are two solutions. Then $ax_1 = b = ax_2$, so $ax_1 = ax_2$. By the left cancellation law, $x_1 = x_2$.

Hence the solution is unique, and $\boxed{x = a^{-1}b}$.

Now consider $ya = b$. Since $b, a^{-1} \in G$, closure gives $ba^{-1} \in G$. Put $y = ba^{-1}$. Then

$$\begin{aligned} ya &= (ba^{-1})a \\ &= b(a^{-1}a) && (\because \text{associativity}) \\ &= be && (\because a^{-1}a = e, \text{ inverse axiom}) \\ &= b && (\because \text{identity axiom}). \end{aligned}$$

Thus $y = ba^{-1}$ is a solution.

If y_1 and y_2 are two solutions, then $y_1a = b = y_2a$, so $y_1a = y_2a$. By the right cancellation law, $y_1 = y_2$. Therefore, the solution is unique and $\boxed{y = ba^{-1}}$.

Other Problems

Problem 2.9. Prove that if $a^2 = e$ for every $a \in G$, then G is abelian. (OR)

Prove that in a group G , if every element has its own inverse, then G is abelian. (OR)

Prove that in a group G , if for every $a \in G$, $a^{-1} = a$, then G is abelian

(Question 10-QB)

Solution

For every $a \in G$, $a^2 = e \implies aa = e \implies a^{-1}(aa) = a^{-1}e \implies (a^{-1}a)a = a^{-1} \implies a = a^{-1}$. Hence a is its own inverse.

Similarly, $b^{-1} = b$ for every $b \in G$.

Now let $a, b \in G$. Since $ab \in G$ by closure, the hypothesis gives $(ab)^2 = e$.

Hence ab is its own inverse: $(ab)^{-1} = ab$. But,

$$\begin{aligned} (ab)^{-1} &= b^{-1}a^{-1} && (\because \text{inverse of a product}) \\ &= ba && (\because a^{-1} = a \text{ and } b^{-1} = b). \end{aligned}$$

Therefore $ab = ba$ for all $a, b \in G$. Hence G is abelian.

Problem 2.10. Show that, for a group G , $(ab)^2 = a^2b^2$ for all $a, b \in G$ if and only if G is abelian.

(Question 7-QB)

Solution

Suppose first that G is abelian. Then $ba = ab$, and hence

$$\begin{aligned} (ab)^2 &= abab & (\because \text{definition of square}) \\ &= aabb & (\because ba = ab, \text{commutativity}) \\ &= a^2b^2 & (\because \text{definition of powers}). \end{aligned}$$

Thus $(ab)^2 = a^2b^2$.

Conversely, suppose that $(ab)^2 = a^2b^2$ for all $a, b \in G$. Then

$$\begin{aligned} abab &= aabb & (\because \text{expanding both sides}) \\ bab &= abb & (\because \text{left cancellation by } a) \\ ba &= ab & (\because \text{right cancellation by } b). \end{aligned}$$

Therefore $ab = ba$ for all $a, b \in G$, so G is abelian.

Problem 2.11. Prove that a group G is abelian if $b^{-1}a^{-1}ba = e$ for all $a, b \in G$.

Solution

Let $a, b \in G$. By hypothesis, $b^{-1}a^{-1}ba = e$. Multiplying both sides on the left by b , we get

$$\begin{aligned} b(b^{-1}a^{-1}ba) &= be \\ (bb^{-1})a^{-1}ba &= b & (\because \text{associativity and identity axiom}) \\ a^{-1}ba &= b & (\because bb^{-1} = e, \text{inverse axiom}). \end{aligned}$$

Now multiply both sides on the left by a :

$$\begin{aligned} a(a^{-1}ba) &= ab \\ (aa^{-1})ba &= ab & (\because \text{associativity}) \\ ba &= ab & (\because aa^{-1} = e \text{ and identity axiom}). \end{aligned}$$

Thus $ab = ba$ for all $a, b \in G$. Hence G is abelian.

Problem 2.12. If G is a group of even order, prove that there exists an element $a \neq e$ such that $a^2 = e$.

Solution

Let $|G| = 2n$ for some positive integer n . We argue by contradiction. Assume that no element other than e satisfies $a = a^{-1}$.

The identity satisfies $e = e^{-1}$. For every non-identity element $a \in G$, our assumption gives $a \neq a^{-1}$.

Also, $a^{-1} \neq e$ whenever $a \neq e$. Therefore the $2n - 1$ non-identity elements can be grouped into disjoint pairs $\{a, a^{-1}\}$, each containing exactly two distinct elements.

Consequently, $2n - 1$ would have to be even. But $2n - 1$ is odd, a contradiction. Hence there exists some $a \neq e$ such that $a = a^{-1}$.

Multiplying by a gives $a^2 = aa^{-1} = e$ by the inverse axiom. Thus there exists $a \neq e$ satisfying $a^2 = e$.

Subgroups

Definition 2.2 (Subgroup). Let $(G, *)$ be a group. A non-empty subset H of G is called a **subgroup** of G if H itself forms a group under the same operation $*$ defined on G .

Example 2.2. Consider the group $(\mathbb{Z}, +)$ and let

$$H = 2\mathbb{Z} = \{\dots, -6, -4, -2, 0, 2, 4, 6, \dots\},$$

then show that H is a subgroup of $(\mathbb{Z}, +)$.

(Question 34-QB)

Solution

We show that H is a subgroup of $(\mathbb{Z}, +)$.

1. Closure:

Let $a, b \in H$. Then $a = 2m, b = 2n$ for some $m, n \in \mathbb{Z}$. Hence $a + b = 2m + 2n = 2(m + n)$. Since $m + n \in \mathbb{Z}$, we have $a + b \in H$. Therefore, H is closed under addition.

2. Associativity:

For $a, b, c \in H$, $(a + b) + c = a + (b + c)$, because addition in $\mathbb{Z}$ is associative.

3. Identity:

Now, for every $a \in H$, $a + 0 = 0 + a = a$. Hence, 0 is the identity element in H .

4. Inverse:

Let $a \in H$. Then $a = 2m$ for some $m \in \mathbb{Z}$. The additive inverse of a is $-a = -2m \in H$.

Hence $H = 2\mathbb{Z}$ is a subgroup of $(\mathbb{Z}, +)$.

Example 2.3. Consider the group $(\mathbb{R} \setminus \{0\}, \cdot)$ under multiplication. Let $H = \{1, -1\}$, then H is a subgroup of $(\mathbb{R} \setminus \{0\}, \cdot)$.

Example 2.4 (Non-example). Consider the group $(\mathbb{Z}, +)$ and the subset

$$H = \{0, 1, 2, 3, \dots\}.$$

Although H is closed under addition and contains the identity element 0, it does not contain the additive inverse of every element.

Hence H is **not** a subgroup of $(\mathbb{Z}, +)$.

Note. Trivial or Improper subgroups. Every group G , with identity element e , has the following subgroups:

1. **The subgroup $\{e\}$.** The set containing only the identity element forms a group under the operation of G , since:

- It is closed: $e * e = e$.
- Associativity is inherited from G .
- e is its identity element.
- The inverse of e is e itself.

Hence, $\{e\} \subseteq G$ is a subgroup of G .

2. **The subgroup G .** Since every set is a subset of itself and G is already a group under its operation, it is a subgroup of itself.

A subgroup H is called a **non-trivial or proper subgroup** if $\{e\} \subsetneq H \subsetneq G$.

Example 2.5. In $(\mathbb{Z}, +)$, the identity element is 0. Thus, $\{0\}$ and $\mathbb{Z}$ are trivial or improper subgroups, while $2\mathbb{Z}$ is a non-trivial or proper subgroup because $\{0\} \subsetneq 2\mathbb{Z} \subsetneq \mathbb{Z}$.

Tests for a subgroup

The two-condition subgroup test

Theorem 2.1. A nonempty subset H of a group G is a subgroup if and only if

- (i) $a, b \in H \implies ab \in H$ and
- (ii) $a \in H \implies a^{-1} \in H$; for all $a, b \in G$.

(Question 30-QB)

Proof

Part-(i). Suppose that H is a subgroup of G . Closure axiom in H gives (i). Inverse axiom in H gives (ii).

Part-(ii). Suppose that (i) and (ii) hold. We verify the group axioms.

1. **Closure:** This is condition (i).
2. **Associativity:** For $a, b, c \in H \implies (ab)c, a(bc) \in H$ by (i) and $H \subseteq G \implies (ab)c = a(bc)$ because the operation is associative in G .
3. **Identity:** Let $h \in H$. By (ii), $h^{-1} \in H$, and by (i), $hh^{-1} \in H \implies hh^{-1} \in G$. Since $hh^{-1} = e$ in G , we obtain $e \in H$.
4. **Inverses:** This is condition (ii).

Thus H is a subgroup of G . □

Note (Two-condition subgroup test for additive group). A nonempty subset H of an additive group $(G, +)$ is a subgroup of G if and only if:

- (i) $a + b \in H$ for all $a, b \in H$;
- (ii) $-a \in H$ for every $a \in H$.

Here $-a$ denotes the additive inverse of a in G .

The one-step subgroup test

Theorem 2.2. A nonempty subset H of a group G is a subgroup if and only if

$$a, b \in H \implies ab^{-1} \in H.$$

(KUD: Jan-2026)(Question 31-QB)

Proof

Part-(i). If H is a subgroup of G and $a, b \in H$, then $b^{-1} \in H$ by the inverse axiom in H , and $ab^{-1} \in H$ by closure axiom.

Part-(ii). Assume that $a, b \in H \implies ab^{-1} \in H$. We verify the group axioms.

1. **Identity:** For $h, h \in H$, applying the condition:

$$hh^{-1} = e \in H \quad (\because H \subseteq G \text{ and inverse axiom in } G).$$

2. **Inverses:** For $a, e \in H$, applying the condition:

$$ea^{-1} \in H \implies a^{-1} \in H.$$

3. **Closure:** For $a, b^{-1} \in H$, applying the condition:

$$a(b^{-1})^{-1} \in H \implies ab \in H.$$

4. **Associativity:** For $a, b, c \in H \implies (ab)c, a(bc) \in H$ by closure axiom and $H \subseteq G \implies (ab)c = a(bc)$ because the operation is associative in G .

Thus H is a subgroup of G . □

Note (One-step subgroup test for additive group). A nonempty subset H of an additive group $(G, +)$ is a subgroup of G if and only if $a - b \in H$ for all $a, b \in H$, where $a - b = a + (-b)$.

Example 2.6. Prove that the set of square roots of unity is a subgroup of the group of fourth roots of unity under multiplication.

(Question 33-QB)

Proof

Let $G = \{1, -1, i, -i\}$ be the group of fourth roots of unity under multiplication, and let $H = \{1, -1\}$ be the set of square roots of unity.

Clearly, $H \subseteq G$ and $H \neq \emptyset$.

Let $a, b \in H$. Since $1^{-1} = 1$ and $(-1)^{-1} = -1$, all possible values of ab^{-1} are:

a	b	ab^{-1}
1	1	1
1	-1	-1
-1	1	-1
-1	-1	1

Thus $ab^{-1} \in \{1, -1\} = H$ for every $a, b \in H$. Therefore, by the one-step subgroup test, H is a subgroup of G . $\square$

Example 2.7 (Example 2.2).**(Question 34-QB)****Solution**

Clearly, $H = 2\mathbb{Z} \subseteq \mathbb{Z}$ and $H \neq \emptyset$.

Let $a, b \in H$. Then $a = 2m$ and $b = 2n$ for some $m, n \in \mathbb{Z}$.

Therefore, $a - b = 2m - 2n = 2(m - n) \in 2\mathbb{Z} = H$, since $m - n \in \mathbb{Z}$.

Thus $a - b \in H$ for every $a, b \in H$. Hence, by the one-step subgroup test for additive groups, H is a subgroup of $(\mathbb{Z}, +)$.

Intersection and union of subgroups**Intersection of two subgroups**

Theorem 2.3. If H_1 and H_2 are two subgroups of G , then $H_1 \cap H_2$ is also a subgroup of G .

(Question 32-QB)**Proof**

Since both subgroups contain the identity of G , $e \in H_1 \cap H_2 \implies H_1 \cap H_2 \neq \emptyset$. If $a, b \in H_1 \cap H_2$, then $a, b \in H_1$ and $a, b \in H_2$. Because H_1 and H_2 are subgroups, their inverse and closure properties give $ab^{-1} \in H_1$ and $ab^{-1} \in H_2$. Hence $ab^{-1} \in H_1 \cap H_2 \implies H_1 \cap H_2$ is a subgroup of G . $\square$

A union need not be a subgroup

Example 2.8. In $(\mathbb{Z}, +)$, consider

$$H_1 = 2\mathbb{Z} = \{\dots, -6, -4, -2, 0, 2, 4, 6, \dots\},$$

$$H_2 = 3\mathbb{Z} = \{\dots, -12, -9, -6, -3, 0, 3, 6, 9, 12, \dots\}.$$

For any positive integer d , the set $d\mathbb{Z}$ is a subgroup of $(\mathbb{Z}, +)$: it contains 0, and $dr - ds = d(r - s) \in d\mathbb{Z}$ for $r, s \in \mathbb{Z}$. This proves that H_1 and H_2 are subgroups using the one-step additive subgroup test.

However, $2, 3 \in H_1 \cup H_2$ while $2 + 3 = 5 \notin H_1 \cup H_2$, since 5 is divisible by neither 2 nor 3. Thus the union fails closure under addition and is not a subgroup.

But, if $H_3 = 4\mathbb{Z} = \{\dots, -8, -4, 0, 4, 8, \dots\}$, then $H_3 \subseteq H_1$, so $H_1 \cup H_3 = H_1$, which is a subgroup.

(Question 32-QB)

Exactly when is the union a subgroup?

Theorem 2.4. If H_1 and H_2 are two subgroups of G , then $H_1 \cup H_2$ is a subgroup of G if and only if $H_1 \subseteq H_2$ or $H_2 \subseteq H_1$.

Proof

Part-(i). If $H_1 \subseteq H_2$, then $H_1 \cup H_2 = H_2$, a subgroup. If $H_2 \subseteq H_1$, then $H_1 \cup H_2 = H_1$, also a subgroup.

Part-(ii). Suppose that $H_1 \cup H_2$ is a subgroup of G . Then we need to prove that $H_1 \subseteq H_2$ or $H_2 \subseteq H_1$.

Assume for a contradiction that $H_1 \not\subseteq H_2$ and $H_2 \not\subseteq H_1$.

$H_1 \not\subseteq H_2 \implies a \in H_1$ and $a \notin H_2$. Similarly, $H_2 \not\subseteq H_1 \implies b \in H_2$ and $b \notin H_1$.

Since $a, b \in H_1 \cup H_2$, closure axiom gives $ab \in H_1 \cup H_2$.

There are two cases: $ab \in H_1$ or $ab \in H_2$.

Case 1: $ab \in H_1$. Since $a^{-1} \in H_1$, closure in H_1 gives $a^{-1}(ab) \in H_1$. But

$$\begin{aligned} a^{-1}(ab) &= (a^{-1}a)b && (\because \text{associativity in the subgroup } H_1) \\ &= eb && (\because \text{inverse axiom in the subgroup } H_1) \\ &= b && (\because \text{identity axiom in the subgroup } H_1.) \end{aligned}$$

Thus $b \in H_1$, contradicting the choice of b .

Case 2: $ab \in H_2$. Since $b^{-1} \in H_2$, closure in H_2 gives $(ab)b^{-1} \in H_2$. But

$$\begin{aligned} (ab)b^{-1} &= a(bb^{-1}) && (\because \text{associativity in the subgroup } H_2) \\ &= ae && (\because \text{inverse axiom in the subgroup } H_2) \\ &= a && (\because \text{identity axiom in the subgroup } H_2.) \end{aligned}$$

Thus $a \in H_2$, contradicting the choice of a . Both cases are impossible. Hence $H_1 \subseteq H_2$ or $H_2 \subseteq H_1$. $\square$

Addition and Multiplication Modulo an Integer

Definition 2.3. Let n be a positive integer. For any integers a and b , the *addition of a and b modulo n* , denoted by $a +_n b$, is the least nonnegative remainder obtained when $a + b$ is divided by n .

✚ Thus, $a +_n b = r$, where $a + b = qn + r$ with $0 \leq r < n$, for some integer q .

Example 2.9.

1. $4 +_5 3 = 2$, since $4 + 3 = 7 = 1 \cdot 5 + 2$.
2. $5 +_6 1 = 0$, since $5 + 1 = 6 = 1 \cdot 6 + 0$.
3. $(-4) +_5 1 = 2$, since $-4 + 1 = -3 = (-1) \cdot 5 + 2$.

Theorem 2.5. For every positive integer m , $(\mathbb{Z}_m = \{0, 1, \dots, m-1\}, +_m)$ is a finite abelian group of order m .

Problem 2.13. Show that $G = \{0, 1, 2, 3, 4\}$ is a finite abelian group under addition modulo 5.

(Question 26-QB)

Solution

Let $G = \{0, 1, 2, 3, 4\}$. The Cayley table (composition table) for addition modulo 5 is

$+_5$	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

1. **Closure.** Every entry in the table belongs to G . Hence $a +_5 b \in G$ for all $a, b \in G$.
2. **Associativity.** Let $a, b, c \in G$. Then by the division algorithm, $(a +_5 b) +_5 c$ is the remainder when $(a + b) + c$ is divided by 5 and $a +_5 (b +_5 c)$ is the remainder when $a + (b + c)$ is divided by 5.
Hence $(a +_5 b) +_5 c = a +_5 (b +_5 c)$.
3. **Identity.** For every $a \in G$, $0 +_5 a = a = a +_5 0$. Thus 0 is the identity element.
4. **Inverses.** From the table, $0 +_5 0 = 0$, $1 +_5 4 = 0 = 4 +_5 1$, $2 +_5 3 = 0 = 3 +_5 2$.
Hence the additive inverses of 0, 1, 2, 3, 4 are 0, 4, 3, 2, 1, respectively.
5. **Commutativity.** The Cayley table is symmetric about its main diagonal.
Therefore, $a +_5 b = b +_5 a$ for all $a, b \in G$.

Hence $(G, +_5)$ is a finite abelian group of order 5.

Definition 2.4. Let n be a positive integer. For any integers a and b , the *multiplication of a and b modulo n* , denoted by $a \times_n b$, is the least nonnegative remainder obtained when ab is divided by n .

✎ Thus, $a \times_n b = r$, where $ab = qn + r$ with $0 \leq r < n$, for some integer q .

Example 2.10.

1. $3 \times_5 4 = 2$, since $3 \cdot 4 = 12 = 2 \cdot 5 + 2$.
2. $2 \times_6 3 = 0$, since $2 \cdot 3 = 6 = 1 \cdot 6 + 0$.
3. $(-2) \times_5 3 = 4$, since $(-2) \cdot 3 = -6 = (-2) \cdot 5 + 4$.

Theorem 2.6. Let p be prime and $G = \{1, 2, \dots, p-1\}$. Then $(G, \times_p)$ is a finite abelian group of order $p-1$.

Problem 2.14. Show that $G = \{2, 4, 6, 8\}$ is a finite abelian group under multiplication modulo 10.

(Question 27-QB)

Solution

Let $G = \{2, 4, 6, 8\}$. The Cayley table (composition table) for multiplication modulo 10 is

$\times_{10}$	2	4	6	8
2	4	8	2	6
4	8	6	4	2
6	2	4	6	8
8	6	2	8	4

1. **Closure.** Every entry in the table belongs to G . Hence $a \times_{10} b \in G$ for all $a, b \in G$.
2. **Associativity.** Let $a, b, c \in G$. Then by the division algorithm, $(a \times_{10} b) \times_{10} c$ is the remainder when $(ab)c$ is divided by 10 and $a \times_{10} (b \times_{10} c)$ is the remainder when $a(bc)$ is divided by 10.
Hence $(a \times_{10} b) \times_{10} c = a \times_{10} (b \times_{10} c)$.
3. **Identity.** From the row and column labelled 6,
we have $6 \times_{10} a = a = a \times_{10} 6$ for every $a \in G$. Thus 6 is the identity element.
4. **Inverses.** From the table, $2 \times_{10} 8 = 6 = 8 \times_{10} 2$, $4 \times_{10} 4 = 6$, $6 \times_{10} 6 = 6$.
Since the identity is 6, the inverses of 2, 4, 6, 8 in G are 8, 4, 6, 2, respectively.
5. **Commutativity.** The Cayley table is symmetric about its main diagonal.
Therefore, $a \times_{10} b = b \times_{10} a$ for all $a, b \in G$.

Hence $(G, \times_{10})$ is a finite abelian group of order 4.

Cyclic Groups and Generators

Example 2.11 (Order of an Element). Let G be a group and $a \in G$. The **order of the element** a is the *smallest positive integer* n such that $a^n = e$, where e is the identity element of G .

It is denoted by $\boxed{o(a) = n}$.

- ✚ If no positive integer n satisfies $a^n = e$, then a is said to have **infinite order**.
- ✚ In additive notation, the order of an element $a \in G$ is the smallest positive integer n such that $na = \underbrace{a + a + \cdots + a}_{n \text{ times}} = 0$.

Example 2.12. In the multiplicative group $\{1, -1\}$ For $a = -1$, $(-1)^1 = -1 \neq 1$ and $(-1)^2 = 1$. Therefore, $\boxed{o(-1) = 2}$.

Example 2.13. In $(\mathbb{Z}_6, +)$. For $a = 2$, $2(2) = 2 +_6 2 = 4$ and $3(2) = 2 +_6 2 +_6 2 = 0$. The smallest positive integer is 3. Therefore, $\boxed{o(2) = 3}$.

Example 2.14. In $(\mathbb{Z}, +)$. The identity is 0. For $a = 3$, $3 + 3 + 3 + \cdots + 3 \neq 0$ for every positive integer number of additions. Hence, $\boxed{o(3) = \infty}$.

Note. In any group G with an identity element e . Since $e^1 = e$, the identity element always has order $\boxed{o(e) = 1}$.

✚ The order of an element is the *number of times we need to combine the element with itself to get the identity for the first time*.

Definition 2.5 (Cyclic Group). For $a \in G$, write $\langle a \rangle = \{a^k : k \in \mathbb{Z}\}$.

A group G is **cyclic** if $G = \langle a \rangle$ for some $a \in G$, such an element a is called a **generator** of G .

✚ In additive notation, $\langle a \rangle = \{ka : k \in \mathbb{Z}\}$.

The powers $\dots, a^{-3}, a^{-2}, a^{-1}, a^0 = e, a, a^2, a^3, \dots$ need not be distinct.

Do the integer powers of an element form a subgroup?

Lemma 2.1. For every $a \in G$, $\langle a \rangle$ is a subgroup of G .

Proof

Let $a \in G$. Clearly, $\langle a \rangle \subseteq G$ by closure axiom and $\langle a \rangle \neq \emptyset$, since $a^0 = e \in \langle a \rangle$. Choose arbitrary $x, y \in \langle a \rangle$. By the definition of $\langle a \rangle$, there exist integers r, s such that $x = a^r$ and $y = a^s$. Therefore

$$\begin{aligned} xy^{-1} &= a^r (a^s)^{-1} && (\because \text{substitution}) \\ &= a^r a^{-s} && (\because \text{inverse law for powers}) \\ &= a^{r-s} && (\because \text{product law for powers}) \\ &\in \langle a \rangle && (\because r-s \in \mathbb{Z}.) \end{aligned}$$

Since x and y were arbitrary, the one-step subgroup test shows that $\langle a \rangle$ is a subgroup of G . $\square$

Is every cyclic group abelian?

Theorem 2.7. *Every cyclic group is abelian.*

Proof

Let $G = \langle a \rangle$ and choose arbitrary $x, y \in G$. By the definition of a cyclic group, there exist integers r, s such that $x = a^r$ and $y = a^s$. Therefore

$$\begin{aligned} xy &= a^r a^s && (\because \text{substitution}) \\ &= a^{r+s} && (\because \text{product law for powers}) \\ &= a^{s+r} && (\because \text{commutativity of integer addition}) \\ &= a^s a^r && (\because \text{product law for power}) \\ &= yx && (\because \text{substitution.}) \end{aligned}$$

Since x and y were arbitrary, G is abelian. $\square$

Does the inverse of a generator also generate the group?

Theorem 2.8. *If $G = \langle a \rangle$, then $G = \langle a^{-1} \rangle$. That is, if a is a generator of a cyclic group G , then a^{-1} is also a generator of G .*

Proof

Let $G = \langle a \rangle$ and let $x \in G$. Then $x = a^r$ for some integer r . Hence

$$x = a^r = (a^{-1})^{-r}.$$

Since $-r$ is an integer, x is an integer power of a^{-1} . Thus every element of G can be expressed as an integer power of a^{-1} . Therefore, a^{-1} also generates G . $\square$

Is every subgroup of a cyclic group cyclic?**Theorem 2.9.** *Every subgroup of a cyclic group is cyclic.*

(KUD: Jan-2026)(Question 36-QB)

Proof

Let $G = \langle a \rangle$ be a cyclic group and let H be a subgroup of G .

If $H = \{e\}$, then $H = \langle e \rangle$, so H is cyclic. Suppose $H \neq \{e\}$. Then H contains an element $a^t \neq e$, where t is a nonzero integer. If $t < 0$, then $a^{-t} = (a^t)^{-1} \in H$. Thus H contains a positive power of a .

Let m be the least positive integer such that $a^m \in H$. We claim that $H = \langle a^m \rangle$.

Let $x \in H$. Since $H \subseteq G = \langle a \rangle$, we have $x = a^t$ for some integer t . By the division algorithm, $t = qm + r$, with $0 \leq r < m$, for some integers q, r . Now

$$\begin{aligned} a^r &= a^{t-qm} && (\because \text{division algorithm}) \\ &= a^t a^{-qm} && (\because \text{product law for powers}) \\ &= x(a^m)^{-q} && (\because \text{substitution and power-of-a-power law}) \\ &\in H && (\because x, a^m \in H \text{ and } H \text{ is a subgroup.}) \end{aligned}$$

If $r > 0$, then $a^r \in H$ with $r < m$, contradicting the choice of m as least positive integer. Therefore $r = 0$, and hence $x = a^t = a^{qm} = (a^m)^q \in \langle a^m \rangle$.

Thus every element $x \in H$ is an integral power of a^m .

Hence $H = \langle a^m \rangle$, so H is cyclic. □

When does a power of a generator also generate the group?**Theorem 2.10.** *Let G be a cyclic group of order n generated by a . Then a^m generates G if and only if m and n are relatively prime.*

(Question 37-QB)

Proof

Since $G = \langle a \rangle$ has order n , the element a has order n . Thus $a^n = e$, where n is the least positive integer.

Suppose that $\gcd(m, n) = 1$, then there exist integers u, v such that $mu + nv = 1$.

Therefore

$$\begin{aligned} a &= a^{mu+nv} && (\because mu + nv = 1) \\ &= a^{mu} a^{nv} && (\because \text{product law for powers}) \\ &= (a^m)^u (a^n)^v && (\because \text{power-of-a-power law}) \\ &= (a^m)^u e && (\because a^n = e) \\ &= (a^m)^u && (\because \text{identity axiom.}) \end{aligned}$$

Thus a is an integer power of a^m . Consequently, every integer power of a is also an integer power of a^m . Since $G = \langle a \rangle$, it follows that $G = \langle a^m \rangle$.

Conversely, suppose that $G = \langle a^m \rangle$. Let $d = \gcd(m, n)$. Since $a \in G$, there exists an integer k such that $a = (a^m)^k$.

Suppose $d > 1$. Then

$$\begin{aligned} a^{n/d} &= \left((a^m)^k \right)^{n/d} && (\because a = (a^m)^k) \\ &= a^{mkn/d} && (\because \text{power-of-a-power law}) \\ &= (a^n)^{(m/d)k} && (\because m/d \in \mathbb{Z}) \\ &= e && (\because a^n = e.) \end{aligned}$$

But $0 < n/d < n$, contradicting the fact that a has order n . Therefore $d = 1$, so m and n are relatively prime. $\square$

Can powers of a generator repeat and how many generators does an infinite cyclic group have?

Theorem 2.11. Let $G = \langle a \rangle$ be an infinite cyclic group. Then

- (i) Distinct integer powers of a are distinct; that is, $a^r = a^s$ implies $r = s$.
- (ii) G has exactly two generators, namely a and a^{-1} .

(Question 38-QB)

Proof

Since $G = \langle a \rangle$ is an infinite cyclic group, no positive integer n can satisfy $a^n = e$.

(i) Suppose $a^r = a^s$ for integers r, s . If $r \neq s$, we may assume $r > s$. Then

$$\begin{aligned} a^{r-s} &= a^r a^{-s} && (\because \text{product law for powers}) \\ &= a^s a^{-s} && (\because a^r = a^s) \\ &= e && (\because \text{inverse law for powers and inverse axiom.}) \end{aligned}$$

Since $r - s > 0$, this contradicts the preceding observation. Hence $r = s$.

(ii) By assumption, a generates G . Also, for every integer r ,

$$a^r = (a^{-1})^{-r}.$$

Thus every element of G is an integer power of a^{-1} , so a^{-1} also generates G .

Now let b be any generator of G . Since $b \in G = \langle a \rangle$, we have $b = a^m$ for some integer m . Since $a \in G = \langle b \rangle$, we have $a = b^k$ for some integer k . Therefore

$$\begin{aligned} a^1 &= b^k && (\because a = b^k) \\ &= (a^m)^k && (\because b = a^m) \\ &= a^{mk} && (\because \text{power-of-a-power law.}) \end{aligned}$$

By part (i), $mk = 1$. Since $m, k \in \mathbb{Z}$, either $m = k = 1$ or $m = k = -1$. Hence

$$b = a \quad \text{or} \quad b = a^{-1}.$$

Moreover, $a \neq a^{-1}$ by part (i), since $1 \neq -1$. Therefore G has exactly two generators, a and a^{-1} . $\square$